Composing Numbers: Part + Part = Whole

Number Bonds & Part-Part-Whole • Standard: CCSS.MATH.1.OA.C.6

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Physically compose two parts on the two rows of the Place Value Tray into a whole and connect to RPN binary addition.

Every addition problem composes two independent parts into a single whole (Part 1 + Part 2 = Whole). The stack makes this structure visible: Part 1 rests on Waiting Shelf Y, Part 2 is built on Workbench X, and [+] joins them.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Two-Row Tray & Digit Blocks)

Hands-On Setup: Place the [4] Digit Block in Row 1 (Part 1) and the [3] Digit Block in Row 0 (Part 2) of the Ones column on your Place Value Tray.

Student Prediction (Before Calculator): Combine both parts in the Ones column: 4 + 3 = 7 (swap to the [7] Digit Block).

3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [4] [ENTER] [3] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[4]	4	Part 1 (4) enters register X
[ENTER]	4	4 copied up to register Y; represents Row 1 (Part 1)
[3]	3	Part 2 (3) enters register X; represents Row 0 (Part 2)
[+]	7	Plus combines Part 1 (Y) + Part 2 (X) = Whole (7) in X!

4. STORY CONTEXT (Number Partners on the Workbench: Part-Part-Whole Bonds & Friends of Ten)

Your brain loves structure: when you place two parts on the top and bottom rows of the Place Value Tray, you can see how they compose a whole! In From Blocks to the Stack, Row 1 of the Place Value Tray sits on Waiting Shelf Y (Part 1), Row 0 sits on Workbench X (Part 2), and pressing [+] snaps both rows together into the Whole.

Core Investigation (Student Warm-Up A):

Place [5] in Row 1 (Part 1) and [4] in Row 0 (Part 2) of your Place Value Tray to find the whole. Write the keystrokes.

Decomposing the Whole: Finding the Missing Part

Developing Number Sense • Standard: CCSS.MATH.1.OA.B.4

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: Finding a missing addend (6 + ? = 10) is the inverse of composing a sum. By placing the known Whole (10) on Shelf Y and the known Part (6) on Workbench X, pressing [-] isolates the unknown part.

Blocks & Prediction: Place [1][0] in Row 1 of the Place Value Tray (Whole = 10) and [6] in Row 0 (Known Part 1 = 6). Prediction: What missing part combines with 6 to make 10? (10 - 6 = 4).

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[1] [0] [ENTER] [6] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1]	1	Enter Whole (10)
[0]	10	10 in register X
[ENTER]	10	Whole (10) lifted to register Y
[6]	6	Known Part 1 (6) enters register X
[-]	4	Minus subtracts known Part from Whole: Missing Part is 4!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [+] immediately after the first number without pressing [ENTER] to separate Part 1 and Part 2.

Hands-On Block Remediation: Point to the two rows of the Place Value Tray: pressing [ENTER] is the physical divider that lifts Row 1 (Part 1) up to Register Y before Row 0 (Part 2) is typed into Register X.

Misconception 2: Adding the whole and the known part when solving for a missing addend (e.g., computing 10 + 6 instead of 10 − 6).

Hands-On Block Remediation: Place [1][0] in Row 1 (Whole) and [6] in Row 0 (Known Part) of the Place Value Tray; show that the Whole (10) goes into Register Y, the known part (6) enters X, and [−] reveals the missing part.

4. EXTENSION MATHEMATICAL INSIGHT

Finding a missing addend (6 + ? = 10) is the inverse of composing a sum. By placing the known Whole (10) on Shelf Y and the known Part (6) on Workbench X, pressing [-] isolates the unknown part.

Part-Part-Whole & Number Bonds on the Place Value Tray

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: HOW TWO-ROW TRAY PARTS FIT THE STACK

- Row 1 (top row) of the Place Value Tray holds Part 1 in register Y. Row 0 (bottom row) holds Part 2 in register X.
- Pressing [+] combines both parts into the Whole in register X.
- To find a missing part: start with the Whole in Y, enter Part 1 in X, and press [-]!

Key Sequence: `4` `ENTER` `3` `+`

Investigation 1: Make 6: Part 1 is 2. What is Part 2? Write the keystrokes.

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 2: Part 1 is 7, Part 2 is 2. What is the Whole?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 3: Three parts make a whole: 2, 3, and 4. Add them on the stack.

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 4: The Whole is 10. Part 1 is 7. What is Part 2?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Unit 02: Part-Part-Whole & Number Bonds on the Place Value Tray — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades K–1

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Place [5] in Row 1 (Part 1) and [4] in Row 0 (Part 2) of your Place Value Tray to find the whole. Write the keystrokes.

Core Solution: 9 ([9] Digit Block). Keystrokes: [5] [ENTER] [4] [+]

Core Check Keys: 5 ENTER 4 +

Extension Challenge (Warm-Up B): The whole is 8. Part 1 is 5. Find the missing Part 2 using the Place Value Tray and calculator.

Extension Solution: Missing Part is 3. Keystrokes: [8] [ENTER] [5] [-]

Ext. Check Keys: 8 ENTER 5 -

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Make 6: Part 1 is 2. What is Part 2? Write the keystrokes.

Expected Answer: 4

SC-6 Verification Keys: 6 ENTER 2 -

Investigation 2: Part 1 is 7, Part 2 is 2. What is the Whole?

Expected Answer: 9

SC-6 Verification Keys: 7 ENTER 2 +

Investigation 3: Three parts make a whole: 2, 3, and 4. Add them on the stack.

Expected Answer: 9

SC-6 Verification Keys: 2 ENTER 3 + 4 +

Investigation 4: The Whole is 10. Part 1 is 7. What is Part 2?

Expected Answer: 3

SC-6 Verification Keys: 1 0 ENTER 7 -

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.